

Miniguide to $\text{\LaTeX}$ equation environments on multiple lines

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In each case, the suggested alternative is bold and underlined>.

1 Multiple equations with multiple labels

- “**align**” (package: `amsmath`).

& used to align lines and to separate columns. Spacing between columns is decided automatically.

$$\begin{array}{ll} 1 + 2 = 3, & 1 + 2 = 3, \\ 400 + 500 = 900, & 400 + 500 = 900. \end{array} \quad \begin{array}{l} (1) \\ (2) \end{array}$$

```
\begin{align}
1 + 2 &= 3, & & 1 + 2 &= 3, \\
400 + 500 &= 900, & & 400 + 500 &= 900.
\end{align}
```

- “**alignat**” (package: `amsmath`).

& used to align lines and to separate columns. Spacing between columns must be explicitly inserted.

$$\begin{array}{ll} 1 + 2 = 3, & 1 + 2 = 3, \\ 400 + 500 = 900, & 400 + 500 = 900. \end{array} \quad \begin{array}{l} (3) \\ (4) \end{array}$$

```
\begin{alignat}
1 + 2 &= 3, & & \quad & 1 + 2 &= 3, \\
400 + 500 &= 900, & & \quad & 400 + 500 &= 900.
\end{alignat}
```

- “**gather**” (package: `amsmath`).

All lines aligned at center.

$$\begin{array}{ll} 1 + 2 = 3, & (5) \\ 400 + 500 = 900. & (6) \end{array}$$

```
\begin{gather}
1 + 2 = 3, \\
400 + 500 = 900.
\end{gather}
```

- “**eqnarray**”.

Generally avoided because of inconsistent spacing and errors in labeling.

$$\begin{array}{ll} 1 + 2 = 3, & (7) \\ 400 + 500 = 900. & (8) \end{array}$$

```
\begin{eqnarray}
1 + 2 = 3, \\
400 + 500 = 900.
\end{eqnarray}
```

2 Multiple equations with sublabels

- “**subequations**” wrapping “**align**” (package: `amsmath`).
& used to align lines and to separate columns.

$$\begin{array}{ll} 1 + 2 = 3, & 1 + 2 = 3, \\ 400 + 500 = 900, & 400 + 500 = 900. \end{array} \quad \begin{array}{l} (9a) \\ (9b) \end{array}$$

```
\begin{subequations}
\begin{align}
1 + 2 &= 3, & & 1 + 2 &= 3, \\
400 + 500 &= 900, & & 400 + 500 &= 900.
\end{align}
\end{subequations}
```

3 Equations on multiple lines with one label

- “**multiline**” (package: `amsmath`).
Aligns lines from left to right.

$$\begin{array}{l} a + b + c + d + e + f + g \\ a + b + c + d + e + f + g \\ a + b + c + d + e + f + g = h. \end{array} \quad (10)$$

```
\begin{multiline}
a + b + c + d + e + f + g \\
a + b + c + d + e + f + g \\
a + b + c + d + e + f + g = h.
\end{multiline}
```

- “**aligned**” (inside `equation`; package: `amsmath`).
& used to align lines and to separate columns. Set vertical alignment of label with [t] (top), [c] (center), and [b] (bottom).

$$\begin{array}{ll} 1 + 2 = 3, & 1 + 2 = 3, \\ 400 + 500 = 900, & 400 + 500 = 900. \end{array} \quad (11)$$

```
\begin{equation}
\begin{aligned}[c]
1 + 2 &= 3, & 1 + 2 &= 3, \\
400 + 500 &= 900, & 400 + 500 &= 900.
\end{aligned}
\end{equation}
```

- “**alignedat**” (inside equation; package: amsmath).
& used to align lines and to separate columns. Spacing between columns must be explicitly inserted.

$$\begin{array}{cc} 1 + 2 = 3, & 1 + 2 = 3, \\ 400 + 500 = 900, & 400 + 500 = 900. \end{array} \quad (12)$$

```
\begin{equation}
\begin{alignedat}{2}
1 + 2 &= 3, & \quad 1 + 2 &= 3, \\
400 + 500 &= 900, & \quad 400 + 500 &= 900.
\end{alignedat}
\end{equation}
```

- “**gathered**” (inside equation; package: amsmath).
All lines aligned at center. Set vertical alignment of label with [t] (top), [c] (center), and [b] (bottom).

$$\begin{array}{c} 1 + 2 = 3, \\ 400 + 500 = 900. \end{array} \quad (13)$$

```
\begin{equation}
\begin{gathered}[c]
1 + 2 = 3, \\
400 + 500 = 900.
\end{gathered}
\end{equation}
```

- “**split**” (inside equation; package: amsmath).
& used only to align lines (single column). Can change side and positions of labels document-wise with `leqno` and `tbtags` options in amsmath. Does not have the unlabeled `split*` version.

$$\begin{array}{c} 1 + 2 = 3, \\ 400 + 500 = 900. \end{array} \quad (14)$$

```
\begin{equation}
\begin{split}
1 + 2 &= 3, \\
400 + 500 &= 900.
\end{split}
\end{equation}
```

4 Braced equations

- “**cases**” (inside equation)

$$\begin{cases} \frac{1+2}{3} = \frac{3}{2}, & \text{math,} \\ 400 + 500 = 900, & \text{math.} \end{cases} \quad (15)$$

```
\begin{equation}
\begin{cases}
\frac{1 + 2}{3} = \frac{3}{2}, & \text{math,} \\
400 + 500 = 900, & \text{math.}
\end{cases}
\end{equation}
```

- “**dcases**” (inside equation; package: mathtools).
All elements are in display style.

$$\begin{cases} \frac{1+2}{2} = \frac{3}{2}, & \text{math,} \\ 400 + 500 = 900, & \text{math.} \end{cases} \quad (16)$$

```
\begin{equation}
\begin{dcases}
\frac{1 + 2}{2} = \frac{3}{2}, & \text{math,} \\
400 + 500 = 900, & \text{math.}
\end{dcases}
\end{equation}
```

5 Braced equations with sublabels

- “**subnumcases**” (package: cases)

$$f(x) = \begin{cases} 1 + 2 = 3, & \text{text,} \\ 400 + 500 = 900, & \text{text.} \end{cases} \quad \begin{array}{l} (17a) \\ (17b) \end{array}$$

```
\begin{subnumcases}{f(x) = \label{eq:whole}}
1 + 2 = 3, & \text{text,} \\
400 + 500 = 900, & \text{text.}
\end{subnumcases}
```

6 Optimization problem

- “**aligned**” + “**dcases**”

$$\begin{array}{ll} \min_x & J(x), \\ \text{s.t.} & \begin{cases} f(x) = 0, \\ g(x) < 0. \end{cases} \end{array} \quad (18)$$

```
\begin{equation}
\begin{aligned}
\min_x & \quad J(x), \\
\text{s.t.} & \quad \begin{dcases}
f(x) = 0, \\
g(x) < 0.
\end{dcases}
\end{aligned}
\end{equation}
```